

Training Adequacy Confounds the Evaluation of Structural Priors in Unrolled Channel Estimation for Rydberg Atomic Receivers

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Abstract—Adding a structural prior inside an unrolled channel estimator usually improves the reported metric, and the improvement is usually credited to the prior. We show that in magnitude-only channel estimation for Rydberg atomic receivers this evidence is not decisive, because a known training pathology manufactures it. A per-sample normalized error averaged over a wide SNR range is dominated by its low-SNR tail; here that amounts to 89.7% of the gradient below 5 dB, leaving the high-SNR regime underfitted, so any mechanism that helps there appears to supply information. We propose a matched-adequacy control: measure the per-bin gradient share, retrain both the structured and the unstructured arm with the imbalance removed, and re-measure. Applied to our own Hankel-structured estimator over three training designs and three seeds each, the structural advantage at $\text{SNR} \geq 5$ dB is +1.286 dB under the conventional loss, +0.020 dB when both arms are retrained on the regime of interest, and −0.177 dB when the loss is corrected instead — positive, absent, and negative with nothing changed but how training distributed its effort. Two corrections that share no machinery agree, and a per-bin sign pattern survives where no scalar does. Correcting the loss is itself a published remedy, and we measure what it is worth here: +2.171 dB at $\text{SNR} \geq 5$ dB over three seeds, improving every one of eighteen bin-by-seed cells including the two the weighting penalizes most. A log-domain loss needing no bin edges is a simpler alternative that is no better. Separately, pilot efficiency and pilot-count generalization are distinct quantities differing by 2.231 dB. Results are simulation only.

Index Terms—Algorithm unrolling, channel estimation, evaluation methodology, phase retrieval, Rydberg atomic receiver, structural priors.

I. INTRODUCTION

UNROLLING an iterative estimator into a trained network [1], [2] has become standard for channel estimation, including for Rydberg atomic receivers, where the magnitude-only readout makes the problem one of phase retrieval [3], [4]. A recurring next step is to add an explicit structural prior — a low-rank projection, a sparsity constraint — inside the unrolled loop, and to report the resulting improvement as evidence that the prior supplies information the network lacks.

This paper argues that such evidence is usually not decisive, and that the reason is a convention so widespread it is rarely stated: the training loss is a per-sample normalized error, averaged over a training set spanning a wide SNR range.

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This is a finding about evaluation practice, and we did not arrive at it as critics. We built a Hankel-structured unrolled estimator, measured an advantage at high SNR, and believed it. The control described in Sec. III was run to characterize that advantage, not to remove it. The first casualty of the protocol we propose was our own method, and we report it because a result that survives an author’s attempt to keep it is worth more than one that was never tested.

Related work on SNR-dependent loss design. The imbalance itself is known and we do not claim it. Wiesmayr *et al.* [5] state it directly for ML-assisted detection and decoding: a system learns one parameter set while operating over a range of SNRs, the training data is sampled from that range, and the aggregate loss is then dominated by the low-SNR samples, because a small relative improvement at low SNR moves the cost further than a large one at high SNR. They propose SNR dewatering as the remedy. Work on deep OFDM channel estimation has separately examined training on SNR-restricted samples, reporting the expected trade of low-noise accuracy against high-noise accuracy. **[TODO: VERIFY CITATION: arXiv:2401.05436, fields unconfirmed]** What we take from that line of work is the mechanism and the remedy. What is not taken from it, and what this paper is about, is the consequence for *attribution*: that a known training defect manufactures apparent evidence for a structural prior, so that crediting the prior without controlling for the defect is unsound. Wiesmayr *et al.* identify the pathology and fix it; they do not ask what it does to the evidence.

Contributions.

- 1) An attribution decomposition of an unrolled estimator’s gain, separating the learned filter, the attention block, and unrolling itself against a matched non-unrolled control.
- 2) The consequence of a known training pathology for *attribution*: per-sample normalized error over mixed SNR concentrates the gradient at low SNR [5], so any mechanism that improves high-SNR behaviour will appear to add information, and a structural prior credited on that evidence is credited for the training’s failure rather than its own contribution. We give a matched-adequacy protocol that separates the two.
- 3) Two properties of the structural contrast that a scalar hides: a per-bin sign pattern that replicates across training designs and seeds, and a dependence of the reported sign on the averaging window.
- 4) Confirmation that the known remedy [5] transfers to

unrolled magnitude-only estimation, gaining +2.171 dB at $\text{SNR} \geq 5$ dB over three seeds — with a gain in every bin of every seed, including the two bins the weighting penalizes most.

- 5) A demonstration that pilot efficiency and pilot-count generalization are distinct and are conflated by a single reported curve.
- 6) A measurement of the same structural prior under three training designs, three seeds each: its advantage is +1.286 dB under the conventional loss, +0.020 dB when both arms are retrained on the regime of interest, and −0.177 dB when the loss is corrected instead.

What the control did. Under the conventional loss the prior is worth +1.286 dB at high SNR, on three seeds, positive under every statistic we tried. Retraining both arms on the high-SNR range leaves +0.020 dB, with one seed of three significantly negative. Correcting the loss instead — leaving the training range untouched — leaves −0.177 dB, negative on every seed. The quantity does not shrink by a factor; it stops keeping its sign. Two corrections that share no machinery agree, which is why we read the training design as the cause rather than an accident of either control.

II. SYSTEM MODEL AND ESTIMATOR

An N -element uniform linear array serves K users over P pilots. With $\mathbf{G} \in \mathbb{C}^{N \times K}$, pilots $\mathbf{S}$, known reference beam $\mathbf{B}$ and noise $\mathbf{W}$, the receiver observes magnitudes only,

$$\mathbf{Z} = |\mathbf{GS} + \mathbf{B} + \mathbf{W}|, \quad \mathbf{W}_{np} \sim \mathcal{CN}(0, \sigma^2). \quad (1)$$

Each user’s channel is a sum of L_k plane waves on the array manifold, with $L_k \sim \mathcal{U}\{3, 7\}$ [6] and angles drawn uniformly on $(-90^\circ, 90^\circ)$ (Table I); this is a geometric ULA model, not a clustered one. The classical baseline is an expectation–maximization Gerchberg–Saxton iteration (EM-GS) [7]; the unrolled estimator (“URformer”) replaces T iterations with T learned layers, each a data-consistency step followed by a learned filter, a gate, and a Transformer block across users [3].

We implement this estimator faithfully on the geometric ULA model of (1) rather than reproducing any published result; the operating point differs. Nothing here should be read as a reproduction or a validation of prior work, nor as a critique of any specific paper. The convention we examine is near-universal, and our own estimator uses it.

The reference curve is genie-aided. Where a reference appears, it is an unstructured least-squares estimate formed from the *true* phase of the noiseless field. It is not a bound: it presumes information no receiver has, and being restricted to unstructured least squares it can be passed by an estimator that exploits channel structure — which we in fact observe below 5 dB in Sec. IV-A.

III. ATTRIBUTION AND THE MATCHED-ADEQUACY PROTOCOL

A. Where the gain comes from

Fig. 1 decomposes the unrolled estimator’s 3.345 dB improvement over EM-GS. The learned per-layer filter, 980 parameters, contributes 0.147 dB; the attention block contributes

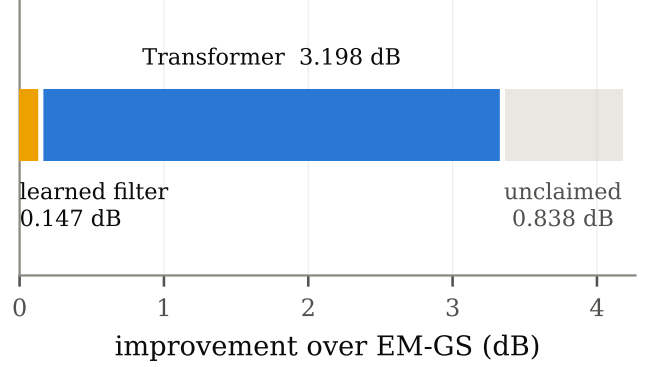


Fig. 1. Attribution of the unrolled estimator’s gain over EM-GS at 80k training samples. “Unclaimed” is the remainder of the genie-aided least-squares headroom, which is a reference and not a bound.

the remaining 3.198 dB, or 96%. The filter’s contribution does not grow with data — it falls slightly, $0.193 \rightarrow 0.147$ dB from 20k to 80k training samples — which is what a never-capacity-limited 980-parameter component should do.

Unrolling is doing real work independently of attention. Against a matched control that applies one Transformer block to a converged EM-GS estimate rather than unrolling, the unrolled estimator gains 0.920 dB, CI $[+0.888, +0.988]$, winning on 86.5% of trials.

B. The confound

The mechanism in this subsection is the one stated in [5]; what is new here is its magnitude in an unrolled magnitude-only estimator, and what it does to attribution.

The training loss is the per-sample normalized error $\|\hat{\mathbf{G}} - \mathbf{G}\|_F^2 / \|\mathbf{G}\|_F^2$, averaged over a batch drawn across the whole SNR range. Because the normalized error of a low-SNR realization is orders of magnitude larger than that of a high-SNR one, this average is dominated by the low-SNR tail. Measured directly on our estimator, 89.7% of the gradient comes from trials below 5 dB, and the per-bin gradient share spans a factor of 30.0 across the range (Fig. 2, upper panel).

The consequence is that the high-SNR regime is systematically underfitted, and *any* mechanism that improves high-SNR behaviour will appear to add information. A structural prior is exactly such a mechanism.

C. The protocol

We propose the following control, which is cheap and which we recommend be run before any structural prior is credited.

- 1) Measure the per-bin gradient share of the training loss. If it is far from uniform, the regime receiving least gradient is underfitted.
- 2) Retrain *both* the structured and unstructured arms with training concentrated on the regime of interest, matching data budget, schedule, seed and initialization.
- 3) Re-measure the contrast. What survives is attributable to the prior; what does not was compensating for training inadequacy.

TABLE I
CONFIGURATION.

Parameter	Value
Array elements N / users K / pilots P	32 / 3 / 20
Paths per user L_k	$\mathcal{U}\{3, 7\}$ [6]
SNR (train and test)	$\mathcal{U}[-10, 20]$ dB
Reference-to-signal ratio	10 dB
Unrolled layers T	10
Trainable parameters	1,586,900
Training samples / epochs	80,000 / 13
Test realizations	2,000 (paired)

Applying it, and reporting what it did to our own method: under the conventional loss the structural advantage over [5, 20] dB is +1.286 dB, averaged over three training seeds. Retraining both arms on [5, 20] dB alone leaves +0.020 dB. Correcting the loss instead, without narrowing the training range, leaves -0.177 dB — the prior now costs accuracy. Sec. IV-E gives the three designs with their seed spreads.

The two corrections share no machinery. One changes which channels the network sees; the other changes how much each channel contributes to the gradient, on an unchanged training set. That they agree is what makes the common cause credible: a single control can always be argued to have broken something incidental, and two unrelated ones are much harder to explain away.

Scope of this negative result. It holds at $N = 32$, $K = 3$, this data budget, and the two pilot counts tested. It is not a general claim that structural priors cannot help unrolled estimators. Indeed the same prior applied to the *classical* estimator, where no training adequacy is at stake, gives a large and stable gain, and under a distribution shift in path richness the structured network degrades least (+0.48 dB against +0.73 dB). The prior buys robustness here; it is the in-distribution accuracy claim that does not survive the control.

IV. NUMERICAL RESULTS

Table I lists the configuration. The reported statistic is the paired per-trial median within an SNR bin, with bootstrap 95% confidence intervals over 2000 paired test realizations.

A. The known remedy transfers to this setting

The confound is a property of the loss, and the remedy for it is already known [5]. This subsection reports what it is worth here. That is a confirmation, not a contribution, and it is placed first only because the rest of the section is easier to read once the loss is fixed.

We weight each sample by a static per-bin factor $w(b) = c/m(b)$, where $m(b)$ is the mean per-sample normalized error in bin b measured once before training, with c set so the weights have unit mean. This requires no schedule, no tuning, and no change to the architecture.

It works as designed. The realized per-bin gradient share spans a factor of 1.9 after balancing against 30.0 before, and the share below 5 dB falls from 0.897 to 0.427 against an ideal of 0.5 — a slight over-correction.

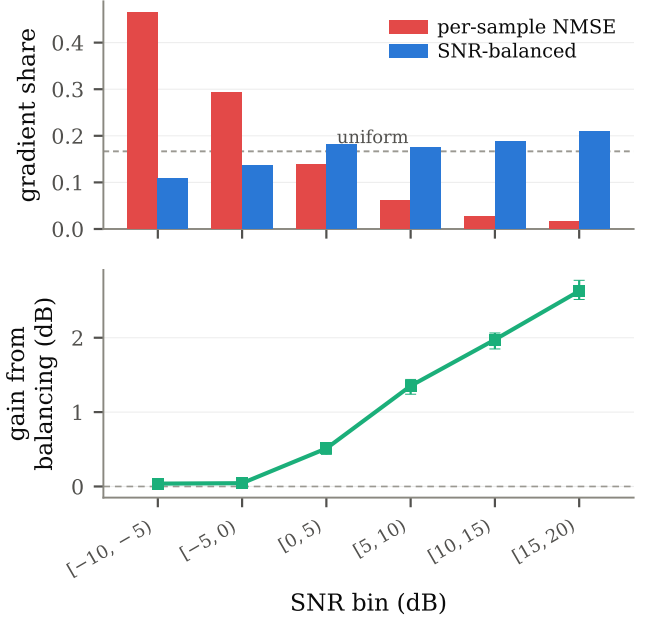


Fig. 2. Cause and effect. Upper: per-bin gradient share under the per-sample normalized loss and under SNR balancing. Lower: the resulting per-bin gain, paired per-trial median with bootstrap 95% CI over test realizations, on the seed for which the prediction was registered.

The estimation result is better than we predicted, and it is measured on three seeds rather than one. Against identically-trained unbalanced baselines, balancing gains +1.902, +2.242 and +2.369 dB at $\text{SNR} \geq 5$ dB, a mean of +2.171 dB. Each seed's bootstrap 95% confidence interval *over test realizations* excludes zero. The spread *over seeds* is a separate quantity and is reported as one: its standard deviation is 0.241 dB, on three seeds and therefore two degrees of freedom, so it is indicative rather than precise. Taking the conservative combination, mean - 2SD = +1.689 dB, the effect is separated from zero across seeds.

We had pre-registered a low-SNR *cost* of -0.10 to -0.40 dB on the reasoning that the lowest bin's weight falls about eighteenfold. There was no cost. On the seed for which the prediction was registered the low-SNR change is +0.135 dB, CI over test realizations [+0.101, +0.172]; across the three seeds the two lowest bins gain +0.052 and +0.202 dB on average. Downweighting the bins that dominated the gradient did not harm them, because they were over-served rather than merely well-served.

The gap to the genie-aided reference at $\text{SNR} \geq 5$ dB falls from +2.897 to +1.000 dB. Below 5 dB the balanced estimator passes that reference by up to 1.99 dB, which is consistent rather than anomalous: the reference is restricted to unstructured least squares, and an estimator that has learned the channel geometry is not so restricted.

B. The correction improves the regime it penalizes

The obvious objection to a reweighting is that it trades one regime for another: whatever the high-SNR bins gain, the

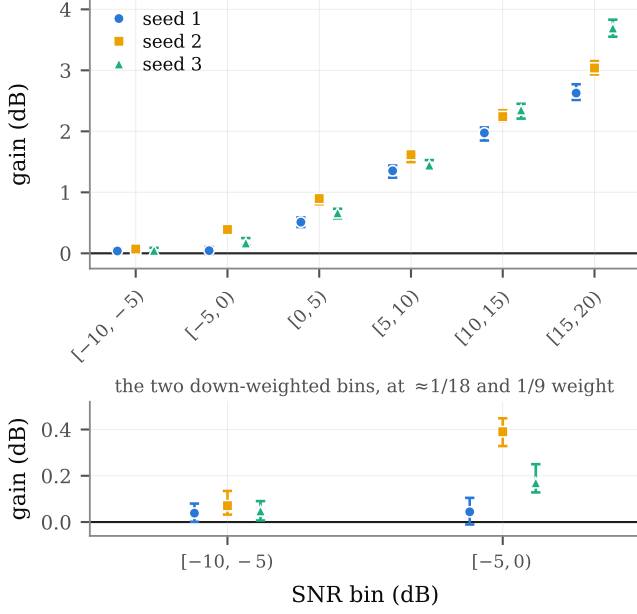


Fig. 3. Every bin, every seed. The gain from SNR balancing, paired per-trial median with bootstrap 95% CI over test realizations, for each of three training seeds. All eighteen cells are positive, including the two lowest bins, whose weights the balancing divides by about 18 and 9.

down-weighted low-SNR bins must have paid. Here they did not, and the check is exhaustive rather than pooled.

Fig. 3 plots every SNR bin for every seed. **All eighteen bin-by-seed cells favour the balanced loss**, and seventeen of the eighteen have a bootstrap 95% confidence interval over test realizations that excludes zero; the single exception is seed 1’s $[-5, 0)$ dB bin at $+0.044$ dB, CI $[-0.011, +0.105]$. The three-seed means rise monotonically across the range, from $+0.052$ dB in $[-10, -5)$ to $+3.120$ dB in $[15, 20)$.

The two lowest bins are the ones that matter for the objection, because the weighting divides their contribution by about 18 and 9 relative to unit mean. They gain anyway. The correction is therefore not a transfer between regimes: it improves the quantity it was weighted away from.

We offer a reading of why, as a hypothesis and not as a result. Under the per-sample loss the gradient was so dominated by the low-SNR tail that the network was underfitted *everywhere*, spending capacity on realizations from which little is recoverable. Rebalancing does not move performance from one regime to another; it stops wasting it. Testing that claim directly would require an ablation we have not run.

C. The structural contrast is not a scalar: a per-bin sign pattern

Reported as one number, the structural gain Δ_H is a weighted average of two opposing effects. Per bin, the prior *costs* accuracy at low SNR and *buys* it at high SNR, and the sign flips exactly once.

Two things about Table II are worth separating.

The ordering replicates and the crossing does not. Every arm we have measured — three seeds of mixed-SNR training,

TABLE II
PER-BIN STRUCTURAL GAIN Δ_H , THREE-SEED MEAN, POSITIVE = THE PRIOR IS BETTER. ALL EIGHTEEN UNDERLYING Δ_H CELLS OF THE MIXED DESIGN — THREE SEEDS BY SIX BINS — HAVE A BOOTSTRAP 95% CI OVER TEST REALIZATIONS EXCLUDING ZERO. THESE ARE A DIFFERENT EIGHTEEN FROM THOSE OF FIG. 3, WHICH ARE THE GAIN FROM BALANCING THE LOSS.

SNR bin (dB)	mixed	SD over seeds	focused
$[-10, -5)$	-0.120	0.036	—
$[-5, 0)$	-0.400	0.103	—
$[0, 5)$	-0.293	0.157	—
$[5, 10)$	+0.457	0.052	-0.173
$[10, 15)$	+1.403	0.089	+0.067
$[15, 20)$	+2.290	0.082	+0.259

three of focused training, and an earlier arm under the balanced loss — places every negative bin before every positive one. On all three mixed seeds the sequence is exactly $- - - + + +$ with all eighteen intervals excluding zero. But the bin at which the sign turns moves by up to two positions between arms. The ordering is the property that survives; the crossing SNR is not, and neither is any scalar summary that depends on where it falls.

The magnitude collapses under matched adequacy while the ordering does not. Comparing the two columns, focused training leaves the shape intact and reduces the size of every bin by roughly an order of magnitude.

D. The averaging window decides the reported sign

Because the per-bin contributions have opposite signs, the number one reports depends on the window the average is taken over — both which bins are inside it and how trials are combined within it. Across three seeds of each design:

- mixed-SNR, restricted to $\text{SNR} \geq 5$: $+1.209, +1.351, +1.298$ dB by paired per-trial median; $+0.723, +0.856, +0.901$ dB by ratio of sums. Positive under both.
- mixed-SNR, pooled over the whole range: $+0.129, +0.225, +0.196$ dB by median; $-0.214, -0.146, -0.141$ dB by ratio of sums. *Sign disagreement.*
- focused training, $\text{SNR} \geq 5$: $+0.078, +0.057, -0.076$ dB by median; $-0.059, -0.077, -0.234$ dB by ratio of sums. Negative under ratio of sums on every seed.

The reason is measurable and we measured it. The paired median weights every trial equally; the ratio of sums weights each trial by its own squared error. Sorting trials by the no-prior arm’s error, the top two deciles carry 51.2% of the total and contribute -0.134 dB, while the remaining eight contribute essentially nothing. That is not a conditioning effect separate from SNR: within an SNR bin the rank correlation between the baseline error and Δ_H is $-0.009, -0.046$ and $+0.147$, against -0.161 overall and $+0.198$ against SNR itself. The sum of squared errors is dominated by low-SNR realizations, which is exactly where Table II says the prior costs accuracy.

It is not the mechanism that produces the same symptom in the classical estimator, where a selection rule abstains on most trials and pins the median at zero. The unrolled arm has

TABLE III
THE STRUCTURAL GAIN Δ_H UNDER THREE TRAINING DESIGNS, THREE SEEDS EACH, ALL ARMS SPECTRALLY INITIALIZED. POSITIVE = THE PRIOR IS BETTER. PER-SEED INTERVALS ARE OVER TEST REALIZATIONS; THE SPREAD IS OVER SEEDS AND IS REPORTED SEPARATELY. THREE SEEDS GIVE TWO DEGREES OF FREEDOM.

Training design	Δ_H	SD_{seed}	$\bar{x} - 2s$	sign	RoS
Conventional loss	+1.286	0.072	+1.142	+++	+++
Focused [5, 20]	+0.020	0.083	-0.147	++-	---
Balanced loss	-0.177	0.150	-0.477	---	---

no such rule: the projection is applied unconditionally, and it moves the estimate on every one of the 10 layers of every one of 2000 trials on every seed, with no trial anywhere near inactive.

E. The structural prior under matched adequacy

Table III is the paper’s central measurement. The prior, the architecture, the data budget, the pilot count and the test set are identical across its three rows. What differs is only how training distributed its effort.

Positive, absent, negative. Under the conventional loss the prior is worth +1.286 dB at $SNR \geq 5$ dB, positive on every seed and under both pooling rules, and separated from zero by twice the across-seed spread. Retrain both arms on the high-SNR range alone and it is +0.020 dB, with one seed of three significantly *negative* and the ratio of sums negative on all three. Correct the loss instead and it is -0.177 dB: negative on every seed, under both pooling rules, with every one of the eighteen per-bin intervals excluding zero.

We state this as a change of category, not as a factor. A ratio between the first and last rows would suggest a quantity that shrank; what the table shows is a quantity that does not keep its sign.

Two interventions, one cause. Focused retraining and loss correction have nothing in common except that each removes the imbalance of Sec. III. One narrows what the network sees; the other reweights an unchanged training set. When two unrelated corrections move a measurement the same way, the common cause is established far more firmly than either could establish it alone — and the second is the more pointed of the two, because it leaves the operating range untouched and so cannot be dismissed as having simply matched training to the test.

What we do not claim. We have not measured Δ_H under the log-domain loss of Sec. IV-F: doing so needs a prior-carrying arm trained under that loss, which we did not train. Three designs are reported, not four. Nor is this a claim that the Hankel prior is useless in general. It holds at $N = 32$, $K = 3$, this data budget and these pilot counts; the same prior applied to the *classical* estimator, where no training adequacy is at stake, gives a large and stable gain.

F. A simpler fix that is no better

Per-bin reweighting needs bin edges, a measurement pass before training to estimate the weights, and a choice of binning. Taking the per-sample error in decibels before averaging

needs none of them: the logarithm divides out the scale factor exactly, per sample. It is the obvious simplification, so we tested it on three seeds.

Against the bin-reweighted arm it gains +0.121 dB at $SNR \geq 5$ dB, with an across-seed SD of 0.131, so the difference is not separated from zero; against the conventional arm both fixes gain about the same, +2.290 dB for the log loss. **The log-domain loss is a simpler alternative that is no better.** We report it because a reader will ask, and because a negative answer to an obvious simplification is worth one paragraph.

It is not, however, better balanced. Its realized share of gradient below 5 dB is 0.317–0.329 against an ideal of 0.5, where per-bin reweighting reaches 0.427; its per-bin span is 3.6 against the reweighted arm’s 1.9. Dividing the scale out exactly does not equalize the gradient, because the derivative of a logarithm is itself inversely proportional to the error.

The imbalance is a property of the loss, not of the weights. The same log-trained networks, scored under the conventional objective, put 0.906–0.911 of the gradient below 5 dB with a span above 40. A model trained to be well balanced still exhibits the pathology the moment it is scored with the objective that causes it — which is the cleanest statement we can give that Sec. III describes the loss and not the network.

G. Pilot efficiency is not pilot-count generalization

A pilot-count curve for a trained estimator can mean two different things: one model trained at some P_0 and evaluated at each P (generalization), or a model trained at each P (efficiency). These are not close. Trained at $P = 10$, the estimator beats the $P = 20$ model evaluated there by +2.231 dB pooled, every bin’s CI excluding zero. At $P = 35$ the corresponding margin is +0.635 dB, so the model generalizes upward in pilot count far better than downward.

At 5 dB the matched-trained estimator reaches -7.52 dB at $P = 10$, within 0.47 dB of the genie-aided reference, while the $P = 20$ model evaluated there is 2.60 dB short of it. The clearest single indication that the curve is tied to its training condition is that the $P = 20$ model is *worse* at $P = 25$ (-11.13 dB) than at $P = 20$ (-11.27 dB): more measurements, worse estimate. Fig. 4 shows all three curves together.

This matters for how such curves are reported. The pilot sweep of [3] is described as “the NMSE performance versus the number of pilot transmissions P , evaluated at a fixed SNR of 5 dB”, and as “crucial for assessing the pilot efficiency of each algorithm”; the text does not state whether the networks were retrained per pilot count. We report that as a fact about what the paper specifies, and as motivation for stating the distinction explicitly — not as a criticism of that work, whose convention is the ordinary one and which our own earlier curves shared.

V. CONCLUSION

A training loss averaged over a wide SNR range is dominated by its low-SNR tail [5]; here that amounts to 89.7%

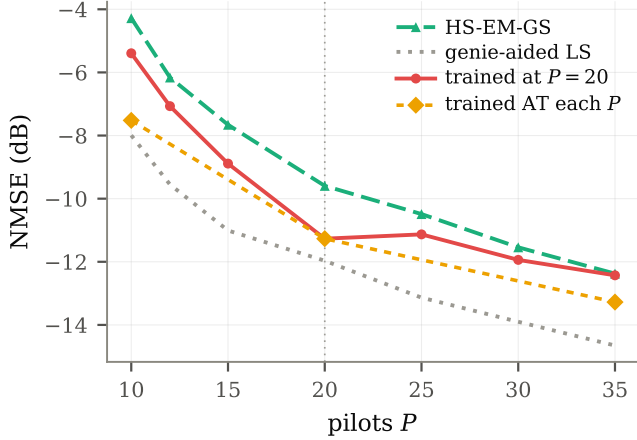


Fig. 4. Pilot efficiency against pilot-count generalization at 5 dB, on identical realizations. The dashed diamonds are separate networks trained at each P ; the solid curve is one network trained at $P = 20$.

of the gradient below 5 dB and leaves the high-SNR regime underfitted. The consequence we draw from it is about evidence rather than about performance: any mechanism that improves high-SNR behaviour will appear to add information it need not have supplied, so a structural prior credited on an uncontrolled comparison is credited for the training’s failure. We recommend the matched-adequacy control be run before a structural prior is credited, and note that it cost us our own result first.

Two properties of the structural contrast survive that control and a single number does not report either: the per-bin sign pattern, which replicates across training designs and seeds while the SNR at which it crosses does not, and the dependence of the reported sign on the averaging window. Applying the known remedy [5] to this setting gains +2.171 dB at $\text{SNR} \geq 5$ dB across three seeds and improves every one of eighteen bin-by-seed cells, including the two the weighting itself penalizes most.

Applied to our own Hankel-structured estimator, the control turns a +1.286 dB advantage into +0.020 dB when both arms are retrained on the regime of interest, and into -0.177 dB when the loss is corrected instead — three designs, three seeds each, nothing changed but how training distributed its effort.

Limitations. All results are simulation only, with no measured hardware and no recorded channel data. They cover the geometric ULA model of (1) at $N = 32$, $K = 3$, one data budget and the pilot counts tested; the negative result about structural priors is scoped to those conditions and is not a general claim. We establish no fundamental bound: the reference curve used throughout is genie-aided, presumes phase no receiver observes, and is passed by our own estimator at low SNR.

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